

ENGMT359 – Lab Task Progress

Overall Task:

1. Download the server file shared with you from your OneDrive. And run the extracted MapSimHD.exe
2. Enter your provided password to get the activation code and use the activation code to start your server.
3. Use the bareMinimum.py from your Moodle page under the Assignment and “lab task instruction” folder.

<https://elearn.waikato.ac.nz/mod/resource/view.php?id=2359929>

4. Use the script as a starting point to get your bot to move and you can edit the below snippet to the existing code to allow for obtaining the exact process's update rate

```
if data.get ("type") == "DATA":
    sensorData = data.get ("sensor", [])
    leftEncoder = data.get ("leftEncoder", 0)
    rightEncoder = data.get ("rightEncoder", 0)
    yawDelta = data.get ("yawDelta", 0)
    dTime = data.get("dt", 0)
    print (leftEncoder,rightEncoder,yawDelta, sensorData, dTime)
```

5. Create a large enough numpy array that can hold the environment data (map) at all the angles possible
6. Use the sensor data from the server (as shown above) and draw your first map exploration region
7. Use the region's boundary (free & obstacle) information to estimate the next point to go and explore.
8. Use a controlled acceleration/deceleration to ensure bot doesn't jump inside an obstacle and gets stuck causing you to rerun the server (max change of 20 pix/s/update_loop or 300 pix/s²).
9. Use A* to do the path plan (this runs every iteration) to estimate the path for the bot to travel and then as the bot moves over each pixel get the new updated sensor information to draw the new information and keep expanding your map.

10. The path finding should keep the bot diameter and clearance of 5 pixels into consideration as you guys did for your A* assessment.
11. Keep exploring the map until no area is left unexplored and you have recreated the unknown map within your script and use this map to demonstrate.
12. As for the marks being awarded, obviously a cleaner map in a shorter time would be marked up higher than noisy map from going way too fast!